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Manual

Energy Demand Forecaster

Time Series Forecasting Tool



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Course of Studies: Business Intelligence and Data Analytics

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Submission date: May 13, 2026

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1 Introduction

Welcome to the **Energy Demand Forecaster** a portable desktop application for predicting electricity consumption from historical meter data.

Purpose

The Energy Demand Forecaster takes a time series of past hourly electricity readings and produces a forecast for a chosen future period from one day up to ninety days. The output is an interactive chart, a set of eight summary statistics, and a downloadable CSV file ready for use in reports or planning tools.

The tool is intended for:

- Energy managers and operations teams planning generation or grid capacity.
- Analysts budgeting future energy costs.
- Anyone who needs a reliable short-to-medium-term demand forecast without writing code or configuring statistical software.

Key features

- **Two forecasting models** — SARIMA (statistical) and LSTM (neural network), selectable per run.
- **Built-in sample dataset** — US PJM East energy demand, ready to use out of the box.
- **Your own data** — upload any CSV file with hourly readings.
- **AI Forecasted Insights** — eight automatic metrics including peak demand, average load, and trend direction.
- **Cross-platform** — identical experience on macOS and Windows.
- **Fully offline** — no internet connection required; your data never leaves your computer.

2 Main Function

The **Energy Demand Forecaster** lets you predict future electricity consumption from historical hourly meter readings. You provide a time series of past energy demand, choose a forecast horizon (e.g. the next 30 days), and the tool produces a chart and a set of summary statistics. No programming or statistics knowledge is required.

Figure 2.1 shows how data moves through the application.

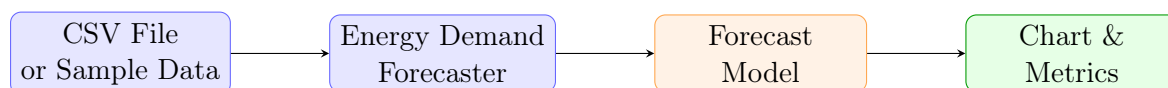


Figure 2.1: Data flow through the Energy Demand Forecaster.



Figure 2.2: Main Dashboard Forecasting PJME Sample

The tool offers two forecasting approaches:

- **SARIMA** — a statistical model. Best for datasets of 60 days or more. A result is ready in about one minute.
- **LSTM** — a neural-network model. Best for datasets of six months or more. Runs instantly on the built-in sample; takes 3–5 minutes on uploaded files.

3 Application Layout

The application runs in your web browser and is divided into two main areas: a **sidebar** on the left for all controls, and a **main panel** on the right for data visualisation and results. Figure 3.1 shows the overall structure.

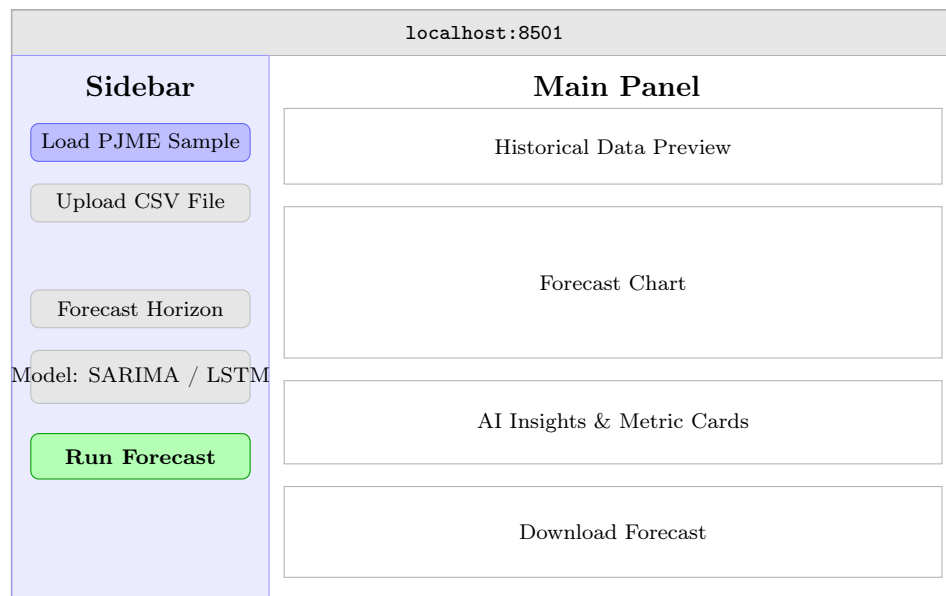


Figure 3.1: Schematic layout of the Energy Demand Forecaster interface.

Main Panel Description

Area	Description
Sidebar	Contains all controls: data loading, column selection, forecast horizon, model choice, and the Run button. All settings must be configured here before running a forecast.
Historical Data Preview	Displays a chart of your loaded dataset immediately after data is uploaded. Use this to confirm the data looks correct before proceeding.
Forecast Chart	After running, shows historical data in blue and predicted values in orange on a single timeline. Hover over the chart to read individual values.
AI Insights & Metric Cards	Eight automatically computed statistics: peak demand, minimum demand, average load, total consumption, demand range, standard deviation, peak hour, and trend direction.
Download Forecast	Saves the full forecast as a CSV file. The file contains timestamps and predicted demand values ready for external use.

Sidebar controls

The sidebar (Figure 3.2) contains every control needed to configure and run a forecast. The controls run top to bottom in the order you use them.

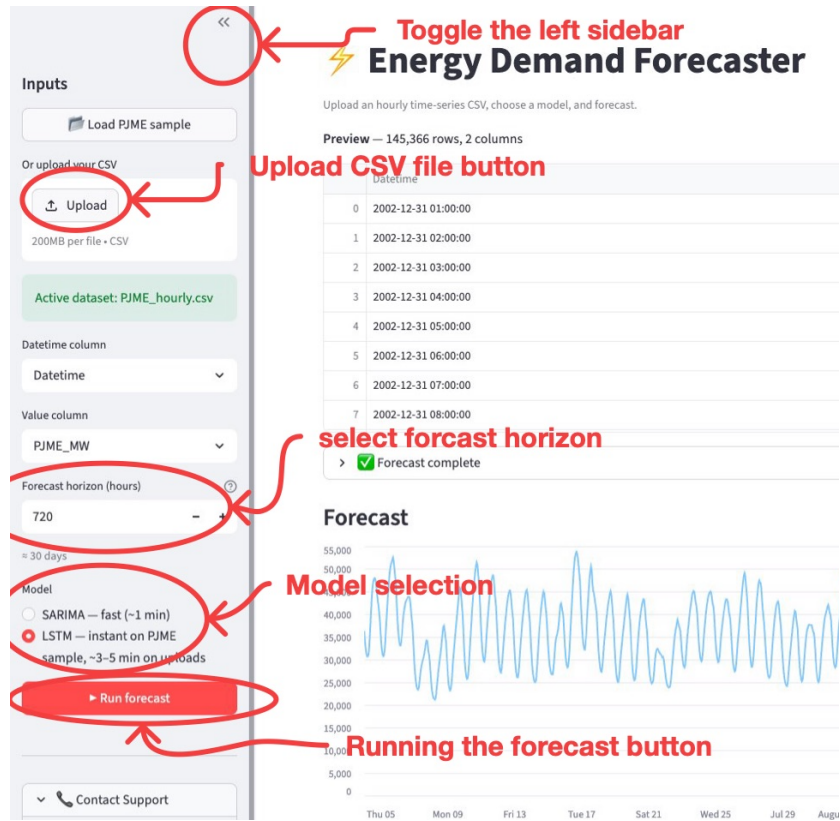


Figure 3.2: Sidebar controls of the Energy Demand Forecaster.

Sidebar Control Description

Control	Function
Load PJME Sample	Loads the built-in US PJM East energy demand dataset (2002–2018). Use this to explore the tool immediately without uploading any file.
Upload CSV File	Opens a file browser to select your own dataset. Must be a CSV with a datetime column and a numeric energy column (see Chapter 6, Step 1 for the required format).
Column selectors	Two drop-down menus that appear after uploading a file. Select which column contains the date/time values and which contains the energy readings.
Forecast Horizon	A slider to set how many hours ahead to forecast. Range: 24 h (1 day) to 2 160 h (90 days). Default: 720 h (30 days).
Model selector	Choose between SARIMA (statistical, ≈ 1 min) and LSTM (neural network, 3–5 min for uploaded files).
Run Forecast	Starts the selected model. The button turns grey while the forecast is running. Do not refresh the page until the result appears.

4 System Requirements

The Energy Demand Forecaster is delivered as a self-contained portable bundle for **macOS** and **Windows**. No separate installation is required.

Table 4.1: Bundle contents

Item	Platform	Description
<code>first_time_setup.command</code>	macOS only	Run once after unpacking to prepare the environment.
<code>run.command</code>	macOS	Starts the application. Double-click to launch.
<code>run.bat</code>	Windows	Starts the application. Double-click to launch. No setup step required.
<code>app/</code>	Both	Application source files (do not modify).
<code>models/</code>	Both	Pre-trained forecast models.
<code>data/</code>	Both	Built-in PJME energy sample dataset.
<code>python/</code>	Windows	Bundled Python runtime (do not modify).

Table 4.2: Minimum system requirements

Requirement	macOS	Windows
Operating system	macOS 12 or later	Windows 10 or later
Processor	Intel or Apple Silicon	64-bit processor
Memory (RAM)	4 GB	4 GB
Free disk space	2 GB	2 GB
Web browser	Any modern browser (Chrome, Firefox, Safari, Edge)	
Internet	Not required	

5 Installation & First Launch

macOS

Figure 5.1 shows the macOS installation steps.

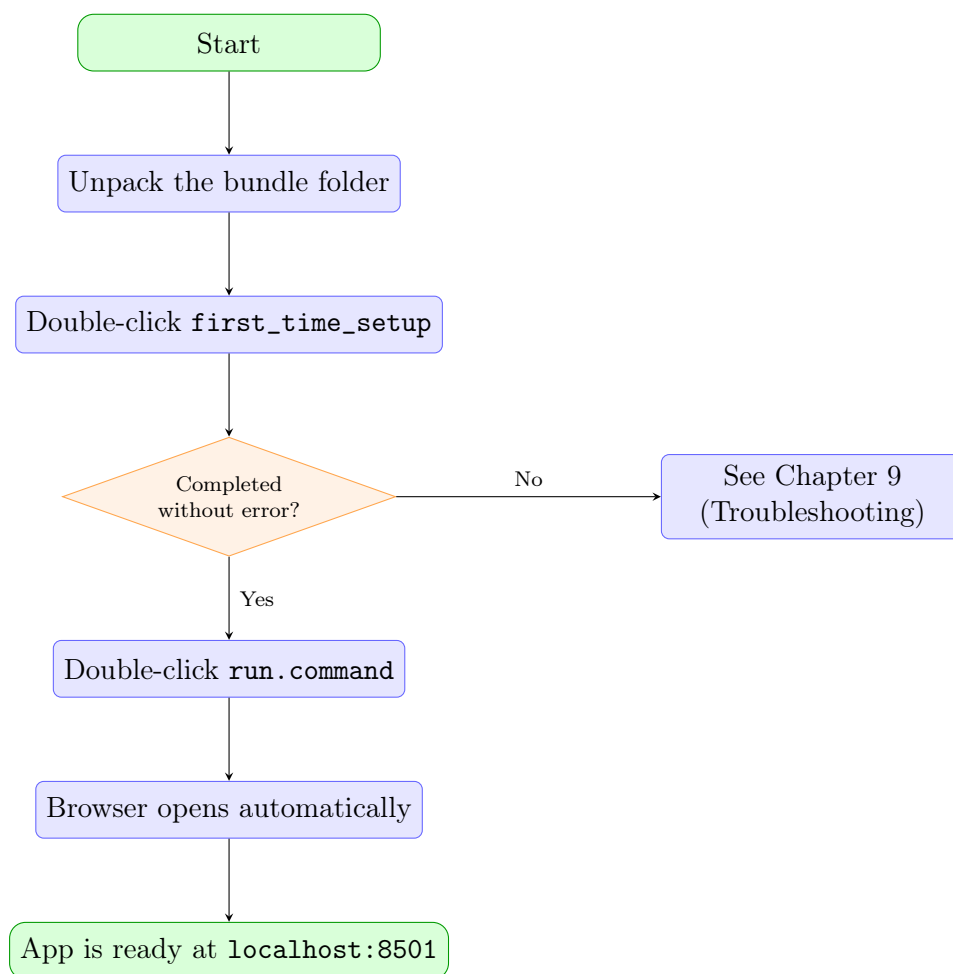


Figure 5.1: macOS installation and launch flowchart.

1. **Download and unpack.** Copy the macOS bundle folder to a location of your choice (e.g. Desktop or Documents). Do not move or rename files inside the folder.

2. **First-time setup (run once only).** Open the bundle folder and double-click `first_time_setup.command`. A terminal window opens and prepares the environment. Wait until the terminal shows *Setup complete* and closes automatically. This step is only needed once per computer.
3. **Launch the application.** Double-click `run.command`. Your default web browser opens and displays the Energy Demand Forecaster. If the browser does not open automatically, type `localhost:8501` in the address bar manually.

Windows

No setup step is required on Windows.

1. **Download and unpack.** Copy the Windows bundle folder to a location of your choice (e.g. Desktop or Documents). Do not move or rename files inside the folder.
2. **Launch the application.** Double-click `run.bat`. A Command Prompt window opens and the application starts. Your default web browser should open automatically after a few seconds. If the browser does not open automatically, type `localhost:8501` in the address bar manually.

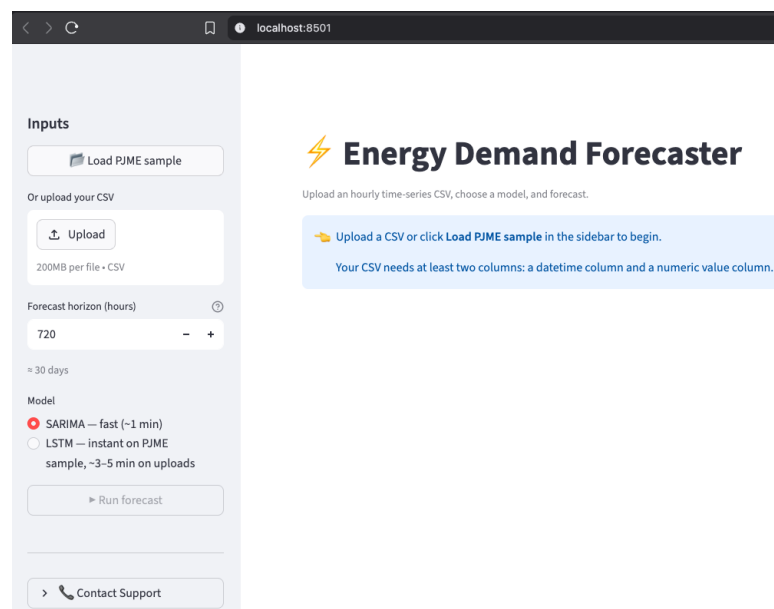


Figure 5.2: Start Up Page

6 Using the Application

Workflow overview

Figure 6.1 shows the five steps to produce a forecast.

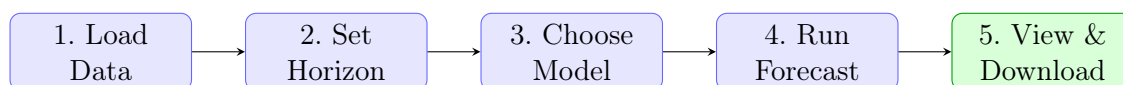


Figure 6.1: Five-step forecasting workflow.

Step 1 — Load data

You can load data in two ways:

- **Built-in sample:** Click **Load PJME Sample** in the sidebar. This loads a ready-to-use dataset of US energy demand. Use this to explore the tool before working with your own data.
- **Your own file:** Click **Upload CSV File** and select a file from your computer. The file must contain at least two columns: a *datetime* column and a numeric *energy demand* column. After uploading, use the two drop-down menus to tell the app which columns to use.

Required data format:

- **Datetime column** — one row per hour, formatted as YYYY-MM-DD HH:MM:SS (e.g. 2024-01-15 14:00:00). Readings must be at *hourly* intervals; irregular or daily data is not supported.
- **Energy column** — numeric values representing power consumption. The built-in sample uses **MW (megawatts)**. Your own data may use any consistent numeric unit (MW, kW, MWh, etc.); the unit label will be shown as-is on the forecast chart.

Energy Demand Forecaster

Upload an hourly time-series CSV, choose a model, and forecast.

Preview — 145,366 rows, 2 columns

	Datetime	PJME_MW
0	2002-12-31 01:00:00	26498
1	2002-12-31 02:00:00	25147
2	2002-12-31 03:00:00	24574
3	2002-12-31 04:00:00	24393
4	2002-12-31 05:00:00	24860
5	2002-12-31 06:00:00	26222
6	2002-12-31 07:00:00	28702
7	2002-12-31 08:00:00	30698

Figure 6.2: Data preview chart after loading the CSV File

Step 2 — Set the forecast horizon

Use the **Forecast Horizon** slider in the sidebar to choose how many hours ahead to forecast. The range is 24 hours (1 day) to 2 160 hours (90 days). The default is 720 hours (30 days).

Step 3 — Choose a model

Figure 6.3 helps you decide which model to use.

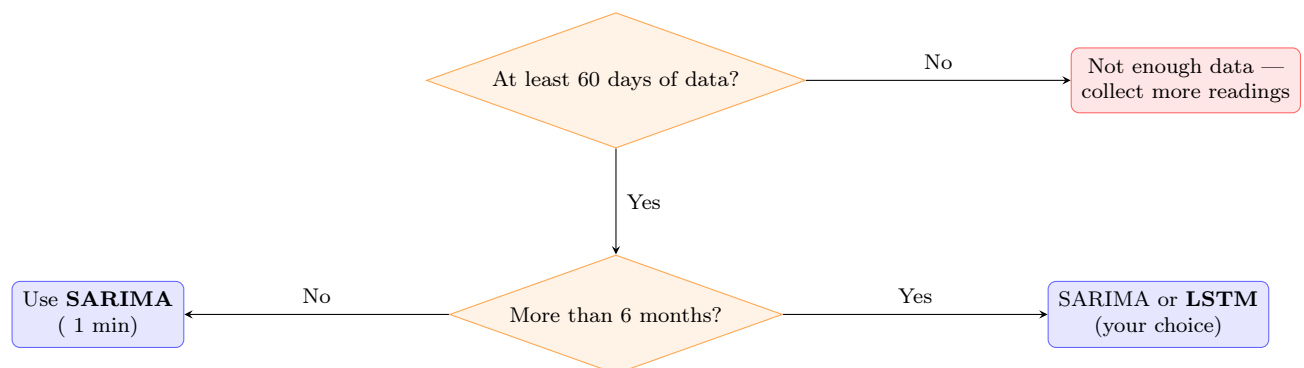


Figure 6.3: Model selection guide.

Select the model using the radio buttons in the sidebar.

Step 4 — Run the forecast

Click the green **Run Forecast** button. A progress indicator appears while the model is working. Do not navigate away from the page during this time.

Step 5 — View and download results

When the forecast is ready, the main panel shows:

- A **forecast chart** with the historical data in blue and the predicted values in orange.
- An **AI Forecasted Insights** panel with eight summary statistics (peak demand, average load, etc.).
- A **Download** button to save the forecast as a CSV file.

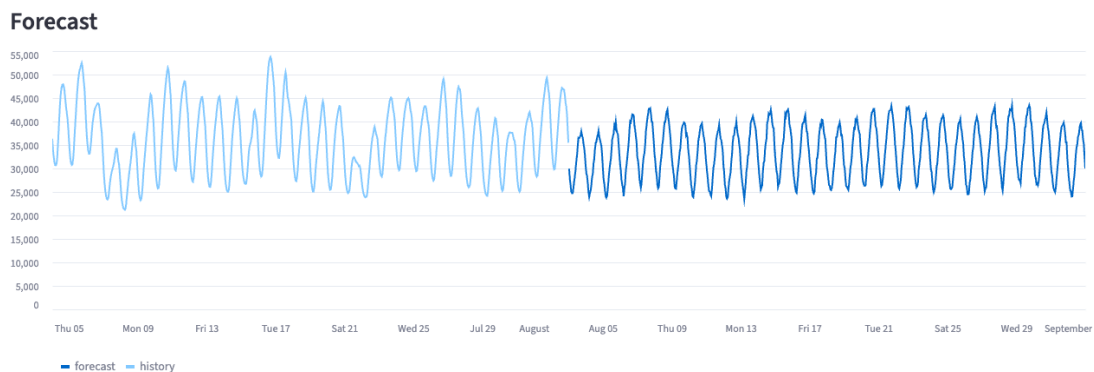


Figure 6.4: Forecast Visualization

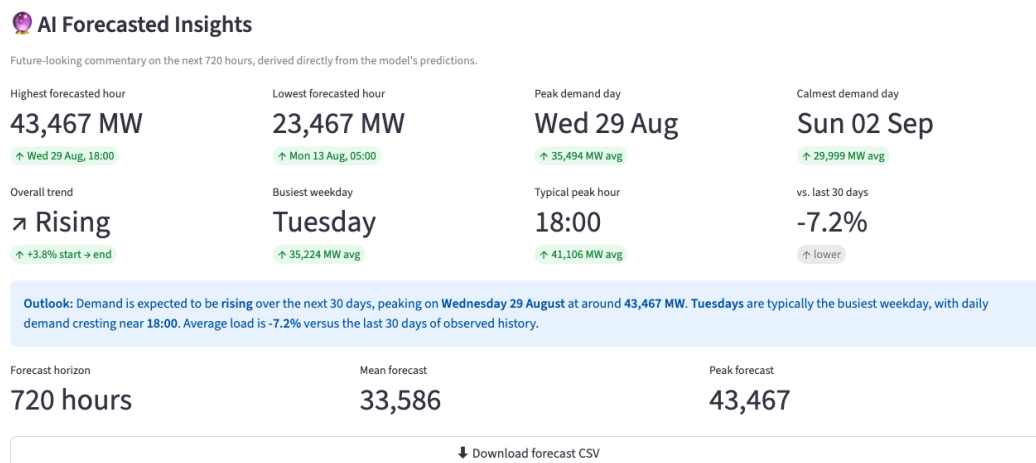


Figure 6.5: AI Forecasted Insights panel and the download button below

7 Maintenance

Keeping data up to date

The built-in PJME sample is a fixed historical dataset and does not update automatically. If you want to forecast using more recent data, export your own data from your energy management system and upload it as a CSV file (see Chapter 6, Step 1).

Disk space

The application bundle uses approximately 1.5 GB of disk space. The bundle folder is located wherever you unpacked it during installation (e.g. **Desktop** or **Documents**). Downloaded forecast results (CSV files) are saved to a folder of your choice and can be deleted at any time without affecting the application.

Updating the application

When a new version is released, download the new bundle and replace the old folder.

- **macOS:** After replacing the folder, double-click `first_time_setup.command` once to re-prepare the environment.
- **Windows:** Simply replace the folder — no setup step is required. Double-click `run.bat` to launch the new version.

Previously downloaded forecast CSV files are not affected by an update.

Restarting the application

If the application stops responding:

- **macOS:** Close the terminal window, then double-click `run.command` again to restart.
- **Windows:** Close the Command Prompt window, then double-click `run.bat` again to restart.

After restarting, navigate to `localhost:8501` in your browser if it does not open automatically.

8 Safety & Usage Guidelines

Before first use

- **No internet connection required.** The application runs entirely on your local computer. Your data is never sent to any external server.
- **Minimum data requirements.** SARIMA requires at least 60 days of hourly readings. LSTM requires at least 6 months of hourly readings. Shorter datasets will produce unreliable forecasts.
- **Do not close the terminal or Command Prompt window** while the app is running. Closing it stops the application immediately.
- **One session at a time.** Do not open two instances of the application simultaneously on the same computer.
- **Forecasts are estimates.** Results depend on the quality and length of the input data. Do not use forecasts as the sole basis for critical decisions without independent verification.

During use

- **Data privacy.** Any CSV file you upload stays on your machine. You are responsible for ensuring that the data you load complies with applicable data protection regulations in your organisation. Do not upload datasets containing personally identifying information.
- **Result interpretation.** Forecast values are statistical estimates, not guarantees. Always cross-check results against operational knowledge before acting on them.
- **File integrity.** Do not modify, delete, or move any files inside the `app/`, `models/`, or `python/` sub-folders. Doing so may cause the application to fail.
- **Single user.** The application is designed for use by one person at a time on one computer. It is not suitable for concurrent access by multiple users over a network.
- **Backup your results.** Back up any important forecast CSV files you download. The application does not store your results internally.

9 Troubleshooting

Table 9.1: Common issues and solutions

Problem	Solution
<i>macOS: Terminal closes immediately after setup</i>	Run <code>first_time_setup.command</code> again. Check that the bundle folder path contains no spaces or special characters.
<i>Browser shows “This site can’t be reached”</i>	The app is still starting. Wait 10 seconds and refresh. If it persists, close the terminal / Command Prompt and relaunch the app.
<i>Browser does not open automatically (common on Windows)</i>	Open your browser manually and type <code>localhost:8501</code> in the address bar. The Command Prompt window must stay open while you use the app.
<i>Upload fails or file not recognised</i>	Ensure the file is a plain CSV (<code>.csv</code>). The first row must be a header. The datetime column must use a standard format, e.g. <code>YYYY-MM-DD HH:MM:SS</code> .
<i>Forecast takes more than 10 minutes</i>	LSTM on large files is slow. Reduce the dataset to the most recent 12 months, or switch to SARIMA.
<i>Forecast chart does not appear</i>	Click Run Forecast again. Ensure both a datetime column and a demand column are selected.
<i>Application crashes mid-forecast</i>	Restart the app and try a shorter horizon or a smaller dataset.

Note. If the problem is not listed above, close the browser tab, close the terminal / Command Prompt, relaunch the app (`run.command` on macOS, `run.bat` on Windows), and try again from the beginning.

10 Add Ons

Planned extensions

The following features are under consideration for future releases:

- **Weather feature integration.** Adding temperature, humidity, and solar irradiance as auxiliary inputs is expected to improve forecast accuracy, particularly for buildings whose demand is strongly driven by heating or cooling loads.
- **Custom model retraining.** A guided workflow will allow users to retrain the LSTM model on their own historical data, enabling the model to adapt to site-specific consumption patterns.
- **Extended forecast horizons.** Support for forecasts beyond 90 days (up to one year) is planned, with automatic aggregation to daily or weekly resolution.
- **PDF report export.** A one-click export will produce a formatted PDF report containing the forecast chart, all AI Forecasted Insights metrics, and model parameters.
- **Multi-meter support.** The ability to load and compare forecasts from multiple meters or sites in a single session.
- **Demand alert thresholds.** Users will be able to define a demand threshold; the tool will flag any forecast hour where predicted consumption exceeds that value.
- **Web deployment.** A server-hosted version of the tool for multi-user access within an organisation, without requiring each user to install a local bundle.

Note. If you have feedback or feature requests, contact the development team through your course supervisor or submit a request via the project repository.